Tejal Anavekar

USC CS grad with data engineering experience at Meta and full-stack experience building responsive web applications and AI products. Strong foundation in system design and solving complex engineering challenges

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Education

University of Southern California — Los Angeles, CA

Master of Science in Computer Science | Graduated May 2026

- Relevant coursework: Analysis of Algorithms, Database management, Applied Natural Language Processing, Information Retrieval, Geospatial Information Management

Fr.C.Rodrigues Institute of Technology — Mumbai, India

Bachelor of Engineering in Information Technology | Graduated June 2024

- Relevant coursework: Operating Systems, Computer Networks, Artificial Intelligence, Big Data, Cloud Computing

Technical Skills

Languages: Python, C, C#, HTML, CSS, SQL, Javascript, Typescript
Engineering Practices: Data Pipelines, REST API Design, System Design, OOP, ETL/ELT
Frameworks: React, Node.js, Express.js, Next.js, ASP.NET Core, REST APIs, Minimal API, Tailwind CSS
Libraries & AI Frameworks: LangChain, PyTorch, Pandas, NumPy
Databases: PostgreSQL, MongoDB, Supabase, Firebase, SQL Server (T-SQL), Redis
Cloud & Tools: AWS (EC2, S3), GCP, Git, Docker, Linux/Unix, Shell Scripting

Experience

Data Engineer Intern — June 2025 – Aug 2025

Meta Platforms — *Burlingame, CA*

- Built and productionized 4 large-scale data pipelines on Meta's internal infrastructure, processing billions of records daily through distributed ingestion to drive real-time analytics and AI readiness dashboards across internal platform operations
- Developed Python-based data validation frameworks enforcing schema integrity, null-safety, and idempotency guarantees
- Optimized DAG-based data workflows through parallel execution with solutions adopted by 3+ cross-functional teams

Software Developer — Sept 2024 – May 2026

USC Institute for Creative Technologies — *Los Angeles, CA*

- Engineered full-stack internal platforms using C# ASP.NET Core Razor Pages and Minimal API endpoints, integrating dependency injection for structured and scalable delivery across internal platforms
- Architected normalized SQL Server schemas, stored procedures, and EF Core data access layers, leveraging LINQ query optimization to accelerate dynamic reporting workflows and reduce Excel report generation time by 40%
- Deployed real-time dashboards to production, implementing end-to-end data validation and monitoring at live environment

Projects

AgentThreads | *Next.js, React, TypeScript, Supabase, Tailwind CSS* | Demo — June 2026 – Present

- Built a responsive social platform for humans and agents, delivering fast web experiences across desktop and mobile
- Implemented Google OAuth and optimistic UI for user interactions, improving responsiveness and seamless user engagement
- Engineered cookie-based SSR authentication and Row-Level Security across 3 core tables, while resolving 2 silent breaking changes from Next.js 16's bleeding-edge API surface

FixIt - E-Learning Platform | *React, Typescript, Node, Express, Tailwind CSS, Supabase, Groq* — May 2026 – Present

- Built an e-learning platform featuring interactive coding sandboxes, in-browser debugging challenges, and adaptive quizzes, delivering engaging hands-on programming practice
- Engineered a schema-validated Zod pipeline enforcing 12+ structured fields, and resolved a bug across 2 OAuth providers
- Iterated on the debugging flow from user feedback, adding concept explanations before the fix step to improve comprehension

Budget Buddy | *MongoDB, Express, React, Node, LangChain, RAG* — Jan 2026 – Present

- Engineered a full-stack personal finance platform with OCR-powered expense receipt scanning and intuitive dashboards, simplifying budgeting and financial tracking
- Implemented JWT- and bcrypt-based authentication, strengthening application security and protecting user financial data
- Built RAG-powered Langchain pipeline delivering personalized financial insights from users' live spending data

Gene Semantic Bridge | *Pytorch, CVAE, BioBert, scRNA-seq, Generative Modeling* — Jan 2026 – May 2026

- Designed a generative AI model integrating language-model embeddings to predict unseen genetic perturbation
- Engineered a unified architecture for single and combinatorial perturbation modeling with improved generalization
- Evaluated on Norman and Replogle OOD benchmarks, achieving R-squared value of 0.996, outperforming baseline models

GeoForecast - Weather App | *MongoDB, Express, React, Node, Swift, GCP* — Aug 2024 – Dec 2024

- Architected a microservices system decoupling geolocation, weather, and user services into independently deployable modules
- Deployed containerized APIs on Google Cloud Platform with autoscaling, improving application scalability and reliability
- Integrated geolocation and weather RESTful APIs to power intuitive dashboards and location-based weather visualizations